Chapter 9
Enhancement of Ecological Connectivity Through UNESCO's Man and Biosphere Reserve: Case of Kinabatangan Landscape, Sabah, Malaysia

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Abstract The Kinabatangan Landscape is located in the eastern part of Sabah, Malaysia, within the greater Kinabatangan basin. It is situated on both sides of the Kinabatangan River, comprising fragmented protected areas (PAs), vast plantation areas, and villages. The area is renowned for its rich biodiversity and unique ecosystem, making it a hotspot for ecotourism. Fragmentation of PAs has occurred due to land clearance for agricultural activities and other land uses starting from the 1980s and 90s. The existing PAs are too small to accommodate sustainable wildlife populations. Hence, the Kinabatangan Landscape needs a platform for stakeholders to work together to achieve common goals, that is, sustainable development. The state government of Sabah approved the proposed nomination of the Kinabatangan Landscape as a UNESCO Man and Biosphere (MAB) Reserve in March 2022. The main purpose of this nomination was to provide a platform for a shared vision for the stakeholders to coexist by encouraging them to synergize their operations toward the UNESCO MAB concept and objectives of sustainable development. The aim was to develop a shared vision as a win–win situation. In preparing the dossier for the nomination, a series of consultations, workshops, meetings, and seminars were held to make the process a win–win both for the people and the nature. Under the UNESCO MAB Programme, involvement in the initiative must be on a voluntary basis. Stakeholders have various objectives for participating in this initiative, which, as much as possible, need to be aligned with government agencies' objectives for integrated management and sustainable development in the Kinabatangan landscape. Therefore, engagement strategies must be tailored to the needs and motivations of the respective stakeholders while also being opportunistic enough to offer benefits from the trade-offs of these conservation opportunities. After more than 2 years of consultation, the perimeter of the biosphere reserve, together with zona-

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tion, was agreed upon by stakeholders. In addition to providing a corridor for wildlife, the nomination of Kinabatangan landscape as a MAB reserve will also have the potential to promote sustainable development by synergizing socioeconomic needs with environmental considerations and promotion of sustainable development.

Keywords Kinabatangan landscape · Sustainable development · MAB reserve · Biodiversity · Tourism

9.1 Introduction

Socio-ecological production landscapes and seascapes (SEPLS) can be located within a larger landscape, and some SEPLS are critically important for the overall sustainability of fauna and flora in the broader landscape. One such area is the Kinabatangan floodplain in Sabah, Malaysia. The Kinabatangan floodplain is home to a number of iconic species in Sabah, such as the orangutan (*Pongo pygmaeus*), North Bornean gibbon (*Hylobates funereus*), proboscis monkey, Bornean banteng (*Bos javanicus lowi*), hornbill, pygmy elephant, numerous birds, and rare flora species of renowned conservation value. This biodiversity has become an attraction for tourists, making the Kinabatangan landscape one of the hotspots for ecotourism in Sabah. The landscape is also fertile because the Kinabatangan River periodically overflows, bringing nutrients to the surrounding area. As a result, most of the areas along the Kinabatangan River have now been developed for agriculture, particularly oil palm. The combination of biodiversity richness and socioeconomic activities has made the Kinabatangan landscape a very good research area in the field of conservation.

Before the 1960s, the Kinabatangan area was covered by thick forest inhabited by wildlife, with a few Orang Sungai settlements scattered along the Kinabatangan River. Orang Sungai, or "people of the river", are one of Sabah's native or Indigenous peoples who predominantly live along rivers on the east coast of Sabah. They depend on the river and the surrounding areas' resources, including wildlife, for their livelihoods. They have a rich cultural heritage, particularly in traditional ecological knowledge, which has developed over a long period in their quest for livelihoods in harmony with river ecosystems. They are involved in various socioeconomic activities, such as agriculture, fishing, swiftlet nest farming, aquaculture, and ecotourism (Ancrenaz et al. 2007).

The timber boom in the 1970s, reaching its peak in the 1990s, altered the Kinabatangan landscape forever (Sabah Biodiversity Centre 2023). Furthermore, the oil palm revolution in Sabah, beginning in the early 1990s, transformed the logged areas into agricultural land. Large swathes of natural forest are now limited to a few fragmented protected areas (PAs) scattered along the Kinabatangan River, which have become refuge areas for the wildlife that originally lived outside them. However, these PAs are too small and fragmented to support all the wildlife within their areas. The Lower Kinabatangan Wildlife Sanctuary (26,103 ha), the largest

among the PAs, is also being encroached upon from time to time by the local communities. Seven more small PAs along the river include the Gomantong, Materis, Bod Tai, Keruak, and Pangi Forest Reserve (1816 ha), the Sg Lokan Forest Reserve (1852 ha), the Pin Supu Forest Reserve (4523 ha), the Gomantong Forest Reserve (3297 ha), the Gologob Forest Reserve (7900 ha), the Balat Damit Wildlife Reserve (4140 ha) and Balat Damit (Extension) Wildlife Reserve (670 ha), and the Trusan Kinabatangan (Extension) (649 ha).

In terms of forest-type distribution within these PAs, there are freshwater swamp forest (19.70%), peat swamp forest (9.24%), beach forest (0.1%), mangrove forest (18.88%), ultramafic forest (0.38%), mixed dipterocarp forest (28.86%), limestone forest (2.32%), seasonal swamp forest (19.89%), and heath forest (0.63%) (Sabah Biodiversity Centre 2023). The diversity of geology and geomorphology directly influences the diversity of fauna and flora in the area (Sabah Biodiversity Centre 2024).

In the last few decades, human–wildlife conflict, especially with elephants and orangutans, has become more rampant. As the PAs are fragmented, the wildlife, particularly the elephants that require large roaming areas, frequently encroaches into oil palm estates and villages. In addition to human–wildlife conflicts, there are also pollution problems stemming from agricultural activities that flow into the river system. Environmental degradation from the discharge of effluents from palm oil mills and nonpoint sources of pollution has a negative impact on aquatic life in the river system, directly affecting the livelihoods of the local communities. There are also complaints from the local communities that they are left behind in socioeconomic development, such as tourism and agriculture-related activities, even though they are local and have been living in the area for generations. Currently, there are about 166,980 people living in Kinabatangan, mostly foreign nationals working in the oil palm sector (Sabah Biodiversity Centre 2024).

At this critical juncture, finding a win–win solution for the coexistence of all stakeholders operating in the Kinabatangan landscape is crucial. It is not merely an option but a necessity to find an appropriate approach to mediate the increasing conflicts and resource clashes within the landscape. With the support of state agencies, several nongovernmental organizations (NGOs) are currently working to promote sustainable development in the landscape. To sustain the NGOs' activities, support is required not only from land users but also from government agencies, local communities, civil society, and international bodies.

Efforts by NGOs to establish ecological connectivity in the early 2000s failed to gain traction due to a lack of support from the government and growers. However, in 2022, the Sabah government agreed to a proposal by the Sabah Biodiversity Centre (SaBC) to nominate the Kinabatangan landscape as a UNESCO Man and Biosphere (MAB) Reserve. UNESCO's MAB is very relevant in this matter because its main objective is promoting sustainable development within the perimeters of PAs. Adopting a MAB approach in this landscape will provide a framework for a common vision of all the actors in the areas. SaBC subsequently appointed HUTAN, a Kinabatangan-based NGO, as a consultant to prepare the dossier for the nomination.

The government's decision to nominate Kinabatangan was formally announced during the State-Level International Biodiversity Conference 2022 held in Kinabatangan. The preparation of the nomination dossier took approximately two-and-a-half years to complete, with the dossier being formally submitted in early July 2024. The dossier, endorsed by both the state and federal governments, is now with UNESCO MAB for consideration.

The MAB concept was chosen as an integrating platform for the Kinabatangan landscape because the approach to sustainable development under it aligns well with the issue of conflict of land uses which we aim to address in this area. Namely, the MAB Programme promotes a holistic approach to addressing the interrelated issues of sustainable conservation, development, and logistical support. By employing the MAB concept, which includes a participatory approach, promotion of research, and zonation, SaBC aspires to integrate and synergize activities in the Kinabatangan landscape toward a shared vision. This vision includes the establishment of ecological corridors, sustainable agriculture, improved livelihoods for local communities, and more inclusive tourism.

As part of the dossier preparation, core, buffer, and transition zones had to be identified and established. Additionally, UNESCO requires that consultations comply with the principle of Free, Prior, and Informed Consent (FPIC).

9.2 Methodology

The proposed vision statement for the nomination outlined the aim to establish a "resilient, interconnected Lower Kinabatangan landscape where people coexist peacefully with nature, integrating flourishing regenerative economies rooted in a shared commitment to sustainable development principles with the preservation of a globally renowned natural and cultural heritage". A dossier was prepared for submission to UNESCO's Man and Biosphere Programme. The dossier included the zonation and draft management plan of the proposed biosphere reserve. Details on the preparation of the dossier are as follows.

9.2.1 Compilation and Documentation Data

Data included recent satellite imagery, locations of the communities, demographic data, land ownership, locations and boundaries of PAs, land uses, administrative data, fauna and flora, and other relevant information. Data on demographic profiles, local issues, and the willingness of stakeholders to participate was also collected. This data was collected through a consultancy process with the stakeholders including local communities, oil palm growers and palm oil mill owners, tour operators, government agencies, NGOs, and civil society. Oil palm growers were divided into three categories—big plantations, medium-sized growers, and smallholders. Consultation with the communities was guided by FPIC principles.

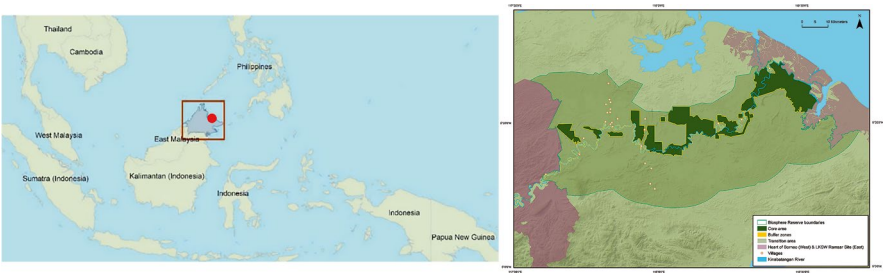

Fig. 9.1 Location maps and the proposed Kinabatangan Biosphere Reserve (KBR). (Source: Based on Final Dossier, Sabah Biodiversity Centre 2024)

Table 9.1 Basic information of the study area

Country/Region	Malaysia
Province	Sabah
District	Kinabatangan
Municipality	Kota Kinabatangan
Size of geographical area (hectare)	413,253 ha
Dominant ethnicity(ies), if appropriate	Orang Sungai
Size of case study/project area (ha)	413,253 ha
Dominant ethnicity in the project area	Orang Sungai
Number of direct beneficiaries (people)	12,700
Number of indirect beneficiaries (people)	154,280
Geographic coordinates (latitude, longitude)	5°23′14.008″ N, 118°4′57.672″ E

9.2.2 Mapping of the Proposed Biosphere Reserve

All data obtained through this data collection, mapping, and engagement process was used to design the management plan and prepare the nomination dossier. This data helped to outline the proposed zonation of the biosphere reserve, including defining the objectives for the zonation of each core area, buffer zone, and transition zone. The proposed Kinabatangan Biosphere Reserve (KBR) is indicated by a red dot (see Fig. 9.1 and Table 9.1).

9.2.3 Stakeholder Engagement

Certain principles were followed in the consultation process with various stakeholders. Consultation procedures with local communities followed these guiding principles: (a) respect communities' rights by getting their consent before the consultation process; (b) be transparent and share the necessary information; (c) conduct the negotiation in good faith by respecting their rights and interests; (d) be sensitive to cultural practices, beliefs, and traditions; (e) promote participatory

engagement by allowing the local communities to express their views freely; and (f) the engagement must be closely monitored. For the oil palm sector, the first step involved identifying the mills and growers within the transition zone. Subsequently, the concept was socialized, and stakeholders were involved in drafting the management plan. For the tourism sector, individual interviews were conducted with all operators, and industry-wide workshops and sharing sessions were also held.

9.3 Findings of Stakeholder Consultations on Proposed Kinabatangan Landscape Biosphere Reserve

Engagement with the stakeholders was quite complex due to the diverse actors involved, each with different expectations and opinions on the proposed nomination. It required perseverance from both SaBC, as the project proponent, and the consultant.

9.3.1 Engagement with Local Communities

No fewer than 25 individual meetings with village heads and district offices have been held. A joint meeting with the communities and district officer was also conducted in May 2023. To ensure that local community leaders could fully participate and understand the objectives of the nomination, meetings were scheduled at their convenience. HUTAN developed a community engagement plan in accordance with the FAO's FPIC Manual and the United Nations Declaration on the Rights of Indigenous Peoples. During the engagement process, the HUTAN team made every effort to ensure that as much as possible all stakeholders were consulted.

Initially, there was a negative connotation associated with the term "reserve" among the local communities, as it could imply government reserve land or the potential gazetting of their land into protected areas, thereby limiting their access to natural resources. However, after detailed explanations, the local communities became supportive of the nomination. They were optimistic about the potential benefits of the nomination for their livelihoods, the preservation of their cultural heritage, and the protection of traditional lands, rights, and practices. They also looked forward to reviving Orang Sungai music and craft-making traditions, which had been declining. Therefore, integrating the cultural heritage of the Orang Sungai is essential for the sustainable management of the Kinabatangan landscape.

9.3.2 Engagement with the Oil Palm Sector

Involvement and commitment from the oil palm sector are critical for successfully achieving the objectives of nominating the Kinabatangan landscape as an MAB Reserve. Oil palm plantations are the major land users in the area, with many operations extending to the fringes of the protected areas (PAs). It is anticipated that approximately 60% of the proposed buffer zone will consist of oil palm plantations. Therefore, it was essential to identify and engage all oil palm growers and mills within the perimeter of the proposed buffer and transition zones. Oil palm operators in the landscape can be categorized as large, medium-sized, and smallholders, each with distinct challenges, perspectives, and approaches. Consequently, the engagement plan must be tailored to these factors. Large operators typically run integrated operations, including mills and possibly refineries, and are required by palm oil certification bodies to establish sustainability units. Engagement with these operators tends to be more straightforward, as they are already familiar with sustainability practices, such as identifying and establishing High Conservation Values (HCVs) within their plantation areas. Additionally, many large operators have adopted Environmental, Social, and Governance (ESG) principles in their business practices, making this nomination a good fit for their sustainability objectives and commitments.

Medium-sized growers own agricultural lands ranging from 40 to 2000 ha. They do not face the stringent requirements imposed directly by palm oil buyers, as their operations are primarily downstream. They sell their oil palm fruits to large operators who own palm oil mills and sometimes refineries. However, the Malaysian Palm Oil Certification Council (MPOCC) mandated that all oil palm operators in Malaysia must be certified by 2019. As a result, all oil palm growers are required to obtain Malaysian Palm Oil Certification to sell their fruits to Malaysian Sustainable Palm Oil (MSPO)- or Roundtable on Sustainable Palm Oil (RSPO)-certified mills.

Engaging with medium-sized growers involves collaborating with regulatory authorities, such as the Malaysia Palm Oil Board (MPOB), the Sabah Jurisdictional Approach for Sustainable Palm Oil (JASPO), and Malaysia's Department of Environment, the Jabatan Alam Sekitar, to demonstrate government commitment to the KBR initiative. Both MPOB and JASPO have endorsed the KBR dossier. By participating in this initiative, medium-sized growers can also receive support from government agencies, such as assistance in flood mitigation efforts, which requires cooperation with government bodies and other estate owners to facilitate debris removal from river systems.

Connecting medium-sized operators to the MAB initiative required significant effort in explaining the long-term sustainability benefits of their operations. For smallholders, motivation to participate in the MAB initiative often comes from the prospect of obtaining alternative or additional livelihood opportunities.

9.3.3 Engagement with the Tourism Sector

The consultant developed an engagement plan for the tourism sector with the objective of increasing the understanding of the nomination process. For tourism operators, interest in participating is driven by opportunities for business growth through branding, promotion, and a regulatory framework for sustainable practices. Interviews were conducted with tour operators, ranging from community-based homestays and bed and breakfasts (B&Bs) to high-end lodges. Additionally, a roundtable discussion was held with relevant government agencies. Accordingly, the Tourism Management Plan was developed and integrated into the KBR management plan. This plan encompasses, among other objectives, more systematic promotion of tourism in the area, products, carrying capacity, local community participation in tourism, and a proposed one-stop information center for MAB. Kinabatangan is a popular area for ecotourism in the region. By integrating Orang Sungai culture and research, tourism has the potential to become an added value for the communities.

9.3.4 Engagement with Government Agencies

Government agencies, such as the Sabah Wildlife Department, Sabah Forestry Department, Department of Drainage and Irrigation, Department of Fisheries, Department of Environment, and many others, will benefit from this initiative. The management objectives of the MAB will become the guiding principles for any proposed development in the area. It will indirectly reduce potential conflicts for any future development initiatives across the landscape.

9.3.5 Nongovernmental Organizations (NGOs)

NGOs play a crucial role as boundary brokers in synergizing activities for sustainable development in the proposed KBR. They are also expected to provide capacity-building for the community, as well as to bridge and communicate the local community's concerns to government agencies and corporate owners. NGOs and the Sabah Biodiversity Centre played an important role as boundary brokers in the negotiation of KBR. A boundary broker must be trusted not only by the government but also by the private sector and local community. Most of the stakeholders have had issues with enforcement agencies in the past such as the Sabah Wildlife and Forestry Departments. NGOs and the Sabah Biodiversity Centre can bridge the trust of all the stakeholders which include state agencies.

9.3.6 Researchers and Academia

Research and education are pillars of the MAB Programme. In order for the proposed MAB reserve to become a laboratory for sustainable development, a research component is a necessity in achieving the goals. As the Kinabatangan landscape is an area rich in biodiversity with human-altered landscapes, it can be an excellent study area for researchers. Research is also needed to identify future requirements of the KBR and to establish facts for the purpose of negotiation with the landowners.

9.3.7 International Partners and Donors

We are aiming to establish a robust model of sustainable development in areas where land use conflicts are occurring. Securing international support is crucial for the successful implementation of the action plan. Therefore, international partners and donors must provide financial contributions to ensure its success.

The proposed core area of the Kinabatangan Biosphere Reserve (KBR) aims to ensure the protected area's integrity by providing optimal protection for ecosystems and biodiversity. This includes creating a safe haven for threatened species and connecting all the protected areas along the Kinabatangan River, linking the Heart of Borneo in the west with the Lower Kinabatangan-Segama Wetland (LKSW) Ramsar site in the east. The PAs are Gomantong, Materis, Bod Tai, Keruak and Pangi Forest Reserve, Sg Lokan Forest Reserve, Pin Supu Forest Reserve, Gomantong, Gologob Forest Reserve, Balat Damit Wildlife Reserve, Balat Damit (Extension) Wildlife Reserve, and Trusan Kinabatangan (Extension). Through this connectivity, we hope to establish a wildlife corridor from the interior of Sabah to the east coast. Due to the complexity of the landscapes, the first 3 months of the initiative were dedicated to socializing the concept with all stakeholders.

The buffer zone area is designed to establish a connection or linkage between the protected areas (PAs) mentioned above. The model will be developed to ensure that only sustainable economic activities with minimal or no impact on the environment are permitted in the core areas. It also aims to protect important species, scenic beauty, and the wilderness of the area, positioning it as a premier global ecotourism hotspot.

The initiative aims to provide a buffer zone with a minimum width of 200 m bordering the PAs. This 200 m width requirement is also extended along the Kinabatangan River. This can be achieved by leveraging the existing mandatory environmental and sustainability standards for agriculture. The buffer zones will include legally constituted riparian reserves along the Kinabatangan River and its tributaries. With a width of 50 m, these reserves will act as a sponge or filter, stabilizing riverbanks and aiding in flood mitigation.

The proposed buffer zone will also encompass riverine villages such as Kg Bilit, Abai, Batu Puteh, Sungai Lokan, Sukau, and Balat. Pressured by human–wildlife

conflicts, some plantation estates have opted to establish conservation areas and wildlife corridors within their plantation areas. This approach will serve as the model for establishing and managing the buffer areas.

The total area of the proposed buffer zone is 12,473 ha, with 54% covered by oil palm plantations, 30% consisting of unprotected riverine forest, 2% as riparian reserve, and 13% designated for other land uses. Through negotiation with local communities, efforts will be made to bypass these villages to mitigate human–wildlife conflicts, although decisions will be made on a case-by-case basis through a consultation process. In the long term, the buffer zone will be promoted for research and monitoring on ecosystem regeneration, with the hope of expanding the buffer zone area if anticipated benefits are realized.

The transition zone of the proposed KBR is a mosaic of land use, predominantly dominated by oil palm plantations, which cover 77% of the zone. This area is home to over 154,200 workers, most of whom are migrants. It also contains 19 villages, predominantly inhabited by the Orang Sungai, with an estimated population of about 9300. In these areas, sustainable management will be promoted by ensuring that both the ecological integrity of the KBR and the socioeconomic well-being of the local people are well maintained. Activities with minimal impact, such as ecotourism and swiftlet farming, will be encouraged.

9.4 Discussion

Sustainable development in the Kinabatangan landscape can only be realized if all actors are involved in the process and share a common vision for the future of the area. Therefore, nominating sites such as the Kinabatangan landscape as an MAB Reserve can serve as a platform to achieve this common vision. However, dealing with complex landscapes where there are clashing land uses, such as conservation, tourism, fishing, and agriculture, poses significant challenges.

The first step is to convince political leaders of the value of the KBR initiative. Before the project commenced, SaBC, as the project proponent, obtained state government endorsement for the initiative. Government commitment is critical in convincing stakeholders, especially players in the palm oil industry, to participate in this initiative.

Second, it is essential to communicate ideas to all parties involved. SaBC, with the support of consultants, designed communication strategies for each sector. It was also necessary to involve the right people for specific tasks. Specialists in local community engagement, palm oil sustainability, and local tourism were brought into the nomination process. These experts were crucial for communicating the initiative's objectives and aligning them with the needs and management objectives of stakeholders to create a win–win situation between conservation and socioeconomic development.

Several sectoral workshops and forums were organized to address current issues and concerns and to explore the benefits of this initiative. A multisectoral workshop

was held to refine the strategies and action plans for a common understanding and vision of the landscape.

For integrated management, we need to consider all aspects—economic, social, political, and ecological—and identify opportunities that may arise from trade-offs. In the Kinabatangan landscape, despite clashing land uses, many opportunities can be leveraged, such as the commitment to ecological passages, the rich cultural heritage of the Orang Sungai, and the potential for tourism. Therefore, we need to identify and exploit these opportunities for the sustainable management of the area (Moon et al. 2014). For example, the community's pride and traditions can be aligned with the conservation objectives of the KBR. Promoting a sense of belonging will enhance their commitment to protecting the fauna and flora in the surrounding environment.

In establishing a shared vision to be translated into the management plan, interventions with local communities, especially the Orang Sungai, must align with their perspectives, identity, and values. It is crucial to evaluate the benefits and economic costs and integrate them from the beginning of the initiative (Adams et al. 2010). For local communities, the main motivation for participating is the opportunity to improve their livelihoods. A significant issue for these communities is poverty due to marginalization in socioeconomic activities in the area. However, any intervention on livelihoods must align with their sociocultural values, which are deeply intertwined with riverine ecosystems (Waylen et al. 2010). The communities' personal ties to the surrounding ecosystems are rooted not only psychologically but also in their social and cultural values (Vining and Tyler 1999).

Recognizing and integrating the cultural values of the Orang Sungai in this landscape are crucial for gaining their support for the new management approach. We must also address other social factors, such as enhancing their sense of belonging to the initiative, building confidence in the government, encouraging membership in local and broader networks, and fostering a willingness to collaborate (Knight et al. 2010). By highlighting the uniqueness of the ecosystems and the richness of species in their areas, we can increase the sense of pride and belonging among local communities and motivate them to cooperate in preserving the area (Drew and Henne 2006). They feel victimized by past experiences when the government alienated land to outside companies or designated it as protected areas without proper consultation. Therefore, in consultations with local communities, we need to address and overcome the mistrust that arose from these past actions.

The nomination process of the MAB Programme, which required consultation based on FPIC (Free, Prior, and Informed Consent) principles, convinced the local communities of the government's sincerity. The guiding principles for consultation were instrumental in securing their commitment. Their participation in the process motivates them to view themselves as part of the solution rather than the problem for the Kinabatangan landscape. To ensure the sustainability of their commitment to the MAB initiative, it is essential for them to collectively agree on activities that directly affect them in the landscape. During the consultation process, they expressed a desire to participate in socioeconomic tourism and be involved in the management of the areas. This aligns well with the MAB objectives of sustainable development

and empowering the local community. Empowering local communities in monitoring and, to some extent, in enforcement will enhance their sense of ownership of the protected areas and surrounding regions. This approach will also lead to better protection of the areas, as they become partners rather than problems or objects of conservation.

Furthermore, for state agencies responsible for managing protected areas (PAs), empowering the local community in monitoring activities can reduce the management costs and shift the balance of monitoring costs to benefits (Walker 2009). Collective choice rules, such as the traditional Tagal system, should be promoted because communities are generally more committed to enforcing rules when they can prevent outsiders from accessing resources they are empowered to manage (Ostrom 2009).

For local communities to participate effectively, their capacity must be enhanced. They will only be able to engage in the initiative if they have capable leaders who can organize them and advocate their interests with government agencies and other stakeholders, such as tour operators, plantation owners, and international funding agencies. In addition to bridging gaps and building trust in the MAB initiative, NGOs can also play a role in enhancing the capacity of local communities.

In dealing with oil palm growers, it is important to clearly communicate the issues, challenges, and available options for the government in addressing land use conflicts in these areas. The oil palm sector is certainly looking for benefits from participating in this initiative. They should understand the current situation faced by the government in managing conflicting land uses in the area and the rationale for designating the Kinabatangan landscape as a biosphere reserve. At the same time, they must realize that other options are still available to the state government, such as imposing more stringent regulations on operations if the MAB initiative fails. Two regulating bodies overseeing oil palm activities in Sabah, namely MPOB and JASPO, have endorsed the initiative by signing the dossier of the nomination. This endorsement aims to ensure that all parties involved view the initiative objectively. However, fairness and openness from the government are essential in gaining the commitment of oil palm growers, as trust is critical for the long-term success of the proposed integrated management of the landscape and must be fostered from the initial stage.

The support and commitment of oil palm growers are vital because more than 60% of the buffer zone and 77% of the transition zone consist of oil palm areas. Therefore, identifying conservation opportunities within these palm oil areas is crucial, as the landscape lacks a natural buffer zone due to most plantation areas being on the fringes of protected areas. During the consultation process, negotiations with oil palm growers have focused on providing wildlife connectivity within the oil palm areas. For example, the MAB reserve initiative can offer plantations the opportunity to leverage their existing environmental and sustainability commitments under MSPO certification. This benefits both the plantation sector and conservation efforts in the landscape. By establishing wildlife corridors within their oil palm plantations, the occurrence of human–wildlife conflicts and damage to palm oil can be reduced. Additionally, wildlife visibility in these corridors may present new

business opportunities for plantation owners. They are also encouraged to consider the potential branding benefits of the KBR if they participate in wildlife conservation efforts.

The tourism sector is a major beneficiary of the MAB Programme initiatives. There is significant potential for branding and promotion through the initiative. The cultural aspects of the Orang Sungai can be a key attraction for tourism if properly designed and aligned with expected visitor arrivals. Therefore, engagement with the Sabah Tourism Board is crucial for promoting the enhanced tourism products of Kinabatangan. To ensure the sustainability of the MAB initiative, it is essential to empower local communities to actively participate in tourism activities. Providing them with a stake in this sector will motivate them to cooperate in protecting the PAs and natural resources as a whole (Ancrenaz et al. 2007). Without their involvement, they may be demotivated to conserve biodiversity because it would not benefit them directly. Thus, in addition to promoting Orang Sungai culture, it is necessary to employ them as tour guides for Kinabatangan. The Sabah Biodiversity Centre, through the proposed MAB Information Centre, should play a critical role in introducing the culture and empowering local communities (Shukla 2004). A Tourism Management Plan for Kinabatangan has been developed and annexed to the Proposed Management Plan for KBR.

Research and education will play a crucial role in ensuring the success of the MAB Programme in Kinabatangan. Data is essential to support the decision-making process, and scientific findings are needed to assess the effectiveness of the action plans developed under the initiative. The Danau Girang Research Centre, located in Kinabatangan, will be pivotal not only in conducting research but also in determining the type of research needed, collecting data, and disseminating results to the management authorities of the MAB reserve and other stakeholders for their capacity-building and effective action (Roux et al. 2006). Local communities, especially the younger generation, need to be given a strong scientific foundation to understand the consequences of not implementing the MAB Programme in the landscape. Integrating modern science with traditional knowledge will help find solutions that are politically, socially, and culturally acceptable to all parties involved (UNU-IAS & IGES 2016). Additionally, the media must actively participate in writing and publishing the success stories from this initiative. This will provide recognition to the actors involved and motivate them to achieve new heights of sustainability.

After consultation with the stakeholders, the total area of the KBR covers approximately 413,866 ha. The zones within the KBR are as follows: (a) core zone—50,950 ha; (b) buffer zone—12,473 ha; and (c) transition zone—350,444 ha. The core zone will provide a safe refuge for wildlife and protect threatened ecosystems. The buffer zone will insulate the protected areas (PAs) from outside threats and serve as a connecting corridor between the PAs. The transition zone is intended to support the conservation goals of the MAB Reserve, promote sustainable development, and create synergy with conservation efforts in the core and buffer zones. Activities in the transition zone impact both core and buffer zones, as well as the livelihoods of the local communities, since discharges from the transition area flow

into the riverine system. There is potential for more plantation estates to join the initiative in the future, especially if the expected benefits, such as achieving sustainability in their operations, are realized.

9.5 Conclusion

We can conclude that UNESCO's MAB Programme is a valuable tool for synergizing activities in areas with overlapping or conflicting land uses. The MAB Programme can serve as a unifying platform for stakeholders to achieve sustainability in their activities within the landscape by fostering a shared vision with interlinked goals and multiple objectives, such as in Kinabatangan (UNU-IAS & IGES 2016). The success of connecting the Ramsar site in the east with the Heart of Borneo in the west under this initiative is due to a good understanding of the problems and interests of the stakeholders. However, it will depend on the commitment of all involved parties. This initiative also will provide an opportunity for co-management, where power in the decision-making process is shared among the government, local communities, private sector, NGOs, researchers, and civil society.

The stakeholders in the areas have overlapping and competing interests, and these naturally will lead to misunderstandings on societal and ecological environments and cause conflicts between parties. Therefore the government, especially SaBC, must have the final say in enforcement and monitoring in co-managed areas (Berkes 2007; Ostrom et al. 1999). Benefits generated by the initiative must be equitably distributed among all stakeholders, particularly the local communities. The management plan must be designed to ensure that all layers of society, especially the poor and underprivileged, are involved in both the initial consultation process and the implementation stage. There should also be no monopoly of benefits by one village, privileged group, or elite.

Connecting the experiences of the stakeholders in pursuing sustainability in the landscape with international conservation discourse will help promote the area to the outside world and recognize their commitments (Stokes et al. 2010). Transparency in management is vital for the success of this initiative. All major decisions in these areas must be discussed by a committee overseeing the management plan, which should include representatives from industry, local communities, and civil society. To ensure sustainability and promote active participation from all stakeholders, a MAB Information Center must be built at a location accessible to everyone. This center will serve as the starting point where visitors can obtain first-hand information on the biodiversity, culture, programs, and activities in the Kinabatangan landscape.

The Sabah Biodiversity Centre plays a critical role in integrating the activities and programs designed under the management plan. In doing so, it must earn the trust of all the stakeholders operating in the Kinabatangan landscape.

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